

Adding, Subtracting, and Multiplying Polynomials

Adding and subtracting polynomials, distributing a monomial, and multiplying binomials. Write every answer in standard form — terms in order of decreasing degree, like terms combined.

- Only **like terms** combine: $3x^2$ and x^2 do, $3x^2$ and $3x$ do not.
- To subtract, distribute the minus sign to *every* term of the second polynomial first, then add.
- Multiplying uses the distributive property: every term of the first factor times every term of the second. Exponents *add*: $x^2 \cdot x^3 = x^5$.

1. Add: $(3x^2 + 5x - 2) + (x^2 - 8x + 7)$

Answer: _____

2. Subtract: $(7x^2 - 4x + 1) - (2x^2 + 6x - 9)$

Answer: _____

3. Multiply: $-3x^2(4x^3 - 2x + 5)$

Answer: _____

4. Multiply: $(x + 6)(x - 4)$

Answer: _____

5. Add, and watch for the missing degrees: $(2x^3 - x + 4) + (5x^2 + x - 9)$

Answer: _____

6. Subtract: $(4x^3 + x^2 - 6) - (4x^3 - 3x^2 + 2x - 6)$

Two of the four terms disappear. Say in one sentence which ones and why.

Answer: _____

7. Multiply: $(2x - 5)(3x + 4)$

Answer: _____

8. Add these two-variable polynomials: $(3x^2y - 5xy + 2y^2) + (x^2y + 5xy - 9y^2)$

Answer: _____

9. Multiply: $4xy^2(2x - 3y)$

Answer: _____

10. A student wrote $(x - 3)^2 = x^2 - 9$. Explain in one sentence why squaring a binomial is not the same as squaring each term, then write the correct expansion of $(x - 3)^2$.

Answer: _____

11. Multiply $(3x - 2)(x + 7)$ and write the product in standard form. Then evaluate your product at $x = 2$. A classmate reports the value -2 ; find the value yourself and say in one sentence what the classmate left out.

Answer: _____

12. Synthesis challenge. Simplify completely and write the result in standard form:

$$(x + 2)(x^2 - 3x + 5) - (x - 1)(x + 4)$$

Multiply both products first, then subtract the second from the first.

Answer: _____